Chapter 25

Resin-bonded Splints and Bridges

"If you cannot do great things, do small things in a great way…"

—Napolean Hill

Mobility of teeth has many causes, including traumatic injury to the face, advanced periodontal disease, habits such as thumb sucking and tongue thrusting and malocclusion. In addition, teeth often need stabilization and retention after orthodontic treatment. In the past, clinical procedures for the stabilization of teeth either involved extensive loss of the tooth structure or were poor in appearance. A conservative and esthetic alternative has been made possible by using resin-bonded splints.

Certain criteria must be met when mobile teeth are splinted. Occlusal adjustment may be necessary initially. The splint should have a hygienic design so that the patient is able to maintain good oral hygiene. It also should allow further diagnostic procedures and treatment, if necessary. The resin-bonded splinting technique satisfies these criteria. Light-cured composites are recommended for splinting because they afford extended working time for placement and contouring.

Periodontally Involved Teeth

Loss of bone support allows movement of teeth, resulting in an increased irritation to the supporting tissues and possible malpositioning of teeth. Stabilizing mobile teeth is a valuable treatment aid before, during and after periodontal therapy. Splinting of teeth aids in occlusal adjustment and tissue healing, thus allowing better evaluation of the progression and prognosis of treatment.

A resin-bonded splint via the acid-etch technique is a conservative and effective method of protecting teeth from further injury by stabilizing them in a favourable occlusal relationship. If the periodontal problem is complicated by missing teeth, a bridge incorporating a splint design is indicated (see section Conservative Bridges).

Techniques for Splinting Anterior Teeth

In short-span segments subject to minimal occlusal forces, a relatively simple technique can be used for splinting periodontally involved teeth. Fig. 25.1A illustrates a maxillary lateral incisor that remains mobile because of insufficient bone support even after occlusal adjustment and elimination of a periodontal pocket. Esthetic recontouring with composite augmentation can be accomplished along with the splinting procedure.

- Anaesthesia generally is not required for a splinting procedure when enamel covers the clinical crown. When root surfaces are exposed and extreme sensitivity exists, however, local anaesthesia is necessary. Teeth are cleaned with a pumice slurry, and the shade of light-cured composite is selected. A cotton roll and retraction cords are used for isolation in this instance.
- With a *coarse, flame-shaped diamond instrument*, enamel on both teeth at the proximal contact area is reduced to produce an interdental space approximately **0.5-mm wide**. This amount of space enhances the strength of the splint by providing more bulk of composite material in the connector between teeth. Other enamel areas of the tooth or teeth that need more contour are prepared by roughening the surface with a coarse diamond instrument.
- Where no enamel is present, such as on the root surface, a mechanical lock is prepared with a No. ¼ round bur in the dentin at the gingivo-axial line angle of the preparation.
- After the prepared tooth surfaces are acid-etched, rinsed and dried, a lightly frosted appearance should be observed on the etched enamel surfaces (Fig. 25.1B).
- The adhesive is applied, lightly blown with air and polymerized. A hand instrument is used to place a small amount of composite material in the gingival area. Additional shaping with a No. 2 explorer reduces the amount of finishing necessary later. It is helpful to add and cure composite in small increments, building from the gingival aspect towards the incisal aspect.



Figure 25.1 Splinting and recontouring a mobile tooth using a light-cured composite. (A) Maxillary right lateral incisor is mobile from lack of bone support. (B) Preparations completed and etched. (C) Splinted and recontoured tooth after 4 years.

- Finishing is accomplished with round and flame-shaped carbide burs, fine diamonds and polishing disks and points. The retraction cord is removed and the occlusion is evaluated to assess centric contacts and functional movements. Instructions on brushing and flossing are reviewed with the patient. The result at 4 years is shown in Fig. 25.1C.

Splinting also can be used when the mandibular incisors are mobile because of severe bone loss. The same general steps are followed as described earlier. If further reinforcement is deemed necessary, however, *a plasma-coated woven polyethylene strip* such as **Ribbond** (Ribbond Inc., Seattle, WA) can be used to strengthen the splint. Additionally, the use of flowable composites greatly facilitates the placement of interproximal composite connectors. Fig. 25.2 illustrates a typical case.

- Following isolation with a rubber dam, small spaces (approximately 0.5 mm in width) are created between teeth with a flame-shaped diamond instrument to enable cross-sectionally strong composite connectors (Fig. 25.2A–C).
- Because a fibre-reinforcing material will be used, the *lingual surfaces to be bonded also should be lightly roughened* with an oval diamond to enhance the resin bonds.
- All interproximal and lingual surfaces to be bonded are etched for 15 s with a phosphoric acid-etching gel (Fig. 25.2D), followed by thorough rinsing and drying.
- Round wooden wedges can be used to stabilize the mobile teeth and to help maintain an open gingival embrasure form.
- To prevent any resin from sticking to the wooden wedges, a light coat of petroleum jelly can be placed on the wedges prior to positioning the wedges interproximally. Bonding agent is applied and cured to all etched surfaces (Fig. 25.2E).
- The interproximal composite connectors are then generated by injecting flowable composite into these areas and shaped (if needed) with a No. 2 explorer (Fig. 25.2F). A small amount of flowable composite is placed onto the lingual surfaces (but not cured) to receive the auxiliary splinting strip.
- An appropriate length of splinting material (polyethylene-coated woven fabric) is cut and first saturated with bonding agent. Then, by using a gloved finger, the strip is pressed into uncured composite and cured initially into place (Fig. 25.2G).
- The bonded strip is then covered incrementally with flowable composite, resulting in a smooth lingual surface (Fig. 25.2H). Facial and incisal embrasures are defined with the finishing burs to enhance esthetics.
- After finishing procedures, the rubber dam is removed and the occlusion is evaluated. The final result is seen in Fig. 25.2I and J.

Stabilization of Teeth After Orthodontic Treatment

After orthodontic treatment, teeth may require stabilization with either fixed or removable appliances. The latter method allows continued minor movements for the final positioning of teeth. When this position is reached, it is better to stabilize teeth with a fixed retainer. Removable retainers tend to irritate soft tissue. Also, they may be damaged, lost or not worn, which usually leads to undesired movement of teeth.

Fig. 25.3A shows a patient with a removable orthodontic retainer. Optimal positioning of teeth has been achieved by orthodontic movement; however, stabilization of teeth is required, and the unattractive spaces caused by undersized maxillary teeth need to be closed (Fig. 25.3B). A carefully planned appointment is required to accomplish the following:

I. remove any fixed orthodontic appliance,
II. add composite to close the diastemas and
III. stabilize teeth with a twisted stainless steel wire and composite.

Technique

- After the orthodontic appliance is removed and routine procedures are followed for closing the diastemas (Fig. 25.3C), the occlusion is examined carefully to determine the best position for locating the twisted wire because it will be placed only on the lingual surfaces.
- A sufficient length of twisted stainless steel wire (i.e. 0.45 mm [0.0175 in.] in diameter) is adapted to the

lingual surface of anterior teeth. A stone cast is helpful for adapting the wire.

- The wire must rest against the lingual surfaces passively without tension or interference with the occlusion. In the mouth, waxed dental tape is used to position the wire against teeth and hold it in place while the occlusal excursions are evaluated. The wire is attached only to the lingual fossa of each tooth.
- After the position of the wire has been determined, it is removed and only the enamel in the fossae (not the marginal ridges or embrasures) is etched, rinsed and dried.



Figure 25.2 Splinting of mobile mandibular incisors reinforced with a plasma-coated, polyethylene-woven strip (Ribbond; Ribbond Inc. Seattle, WA). (A and B) Facial and lingual preoperative views of mobile mandibular incisors that need splinting. (C) Preparation consists of roughening proximal surfaces and creating slight interdental spaces to provide bulk to the connector areas of the composite splint. (D) All interproximal and lingual surfaces to be bonded are etched with a phosphoric acid gel. (E) Teeth are stabilized with wooden wedges, and a bonding agent is applied. (F) Interproximal composite connectors are generated by injecting flowable composite. (G) A fibre-reinforcing strip is pressed into the uncured composite on lingual with a gloved finger. (H) The bonded strip is covered incrementally with flowable composite. (I and J) Completed fibre-reinforced composite-bonded periodontal splint seen from facial and lingual views.

(*Continued*)



Figure 25.2 (*cont.*)

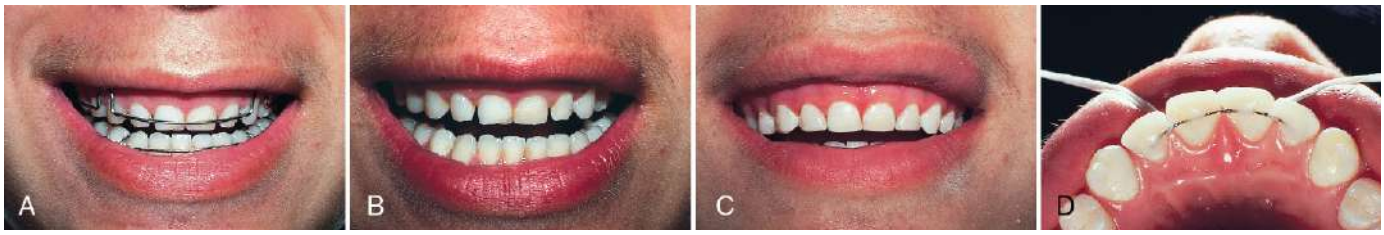


Figure 25.3 Stabilizing teeth after orthodontic treatment. (A) Patient with existing removable retainer. (B) Residual spaces resulting from undersized teeth. (C) Closure of spaces with composite additions is completed. (D) Orthodontic wire is held in position with dental tape and bonded into place with composite.

- Light-cured composite is best used for attaching the fixed wire splint. The wire is repositioned and held in place with a dental tape, while a sparing amount of resin-bonding agent is applied and lightly blown with air.
- After polymerization of the adhesive, a small amount of composite material is placed to encompass the wire in each fossa and bond it to the enamel. The operator must be careful not to involve the proximal surfaces (Fig. 25.3D). After polymerization of composite, the occlusion is evaluated and adjusted, as needed, for proper centric contacts and functional movements.

This unique splint allows some physiologic movement of teeth, yet it holds them in the correct position. The splint should remain in place for *at least 6 months* to ensure stabilization. Longer retention may be necessary, depending on the individual situation and recommendations of the orthodontist.

Avulsed or Partially Avulsed Teeth

Facial injuries often involve the hard and soft tissues of the mouth. The damage may range from lacerations of soft tissue to fractures of teeth and alveolar bone. Partial or complete avulsion of teeth can occur. Maxillary central incisors are involved more often, than are other teeth. A thorough clinical examination of soft tissue, lips, tongue and cheeks should be made to locate lacerations and embedded tooth fragments and debris. Radiographic examination is necessary to diagnose deeply embedded fragments or root fractures.

Treatment of soft tissue lacerations should include lavage, conservative debridement and suturing. Consultation with or referral to an oral surgeon may be necessary. A partially avulsed tooth is repositioned digitally and may or may not need splinting.

Traumatically avulsed teeth that are reimplanted immediately or within 30 min have a good prognosis for being retained.[1,2] After 30 min, the success rate declines rapidly. The avulsed tooth should be repositioned as soon as possible. In the interim, it should be placed in a moist environment such as saliva (i.e. held in the cheek or under the tongue), tooth-saver solution (i.e. Hank' s balanced solution), milk, saline or a wet towel. The replacement of avulsed teeth has immediate psychologic value and maintains the natural space in the event that a fixed prosthesis is required later.

Technique

- The maxillary right incisors that were completely avulsed in an accident (Fig. 25.4A) are repositioned immediately. After the teeth are repositioned, radiographs reveal that no other complications exist. Isolation with cotton rolls or gauze is preferable to the use of a rubber dam, which could cause malpositioning of the loose teeth. The occlusion should be evaluated to ensure that the teeth are properly positioned.
- The facial surfaces of the crowns are quickly cleaned with hydrogen peroxide, rinsed and dried by blotting with a gauze or cotton roll or by lightly blowing with air. The dentist should avoid blowing air into areas of avulsion or deep wounds to prevent air emboli.
- If a crown is fractured, deeply exposed dentin may need to be protected with a liner or base material.
- A twisted orthodontic wire (0.49 mm [0.0195 in.]) must be long enough to cover the facial (or lingual) surfaces of enough teeth to stabilize the loose teeth. The wire is adapted and the ends rounded to prevent irritation to soft tissue. In an emergency, a disinfected paper clip can be used as a temporary splint.
- No preparation of the enamel surface is necessary other than that provided by acid-etching. The middle third of the facial surfaces are etched, rinsed and dried of all visible moisture.
- Drying should be accomplished by blotting with a gauze or cotton roll and a light stream of air. Self-cured or light-cured composite may be used. The wire is positioned and held lightly in place, and the ends are attached with composite material (Fig. 25.4B).
- Light pressure is applied to the repositioned teeth as the facial surfaces are bonded to the wire in succession. Care is exercised not to allow composite to flow into the proximal areas. When the teeth are stabilized, any fractured areas can be conservatively repaired by the acid-etch and resin-bond technique.
- Finishing is accomplished by a flame-shaped carbide finishing bur and abrasive disks. The occlusion is evaluated carefully to ensure that no premature contacts exist.
- The patient is advised to maintain gentle care of the involved teeth. Antibiotic therapy may be required if the alveolar bone is fractured or a significant soft tissue damage has occurred. Tetanus shots or boosters are advised, if indicated by the nature of the accident; the patient's physician should be contacted about this.
- Appointments are made for *follow-up examinations on a weekly basis* for the first month.
- The patient is warned about symptoms of pulpal necrosis and advised to call if a problem develops. If root canal therapy is required, it is better accomplished with the splint in position.
- Removal of the splint is accomplished in *4–8 weeks* provided that recall visits have shown normal pulp test results and the teeth are asymptomatic. The wire is sectioned, and the resin material is removed with a flame-shaped, carbide finishing bur at high speed with air-water spray and a light intermittent application. Abrasive disks are used to polish the teeth to a high lustre.

Conservative Bridges

In selected cases, conservative bridges can be made by bonding a pontic to the adjacent natural teeth. These conservative bridges are classified according to the type of pontic:

I. natural tooth pontic,
II. denture tooth pontic,

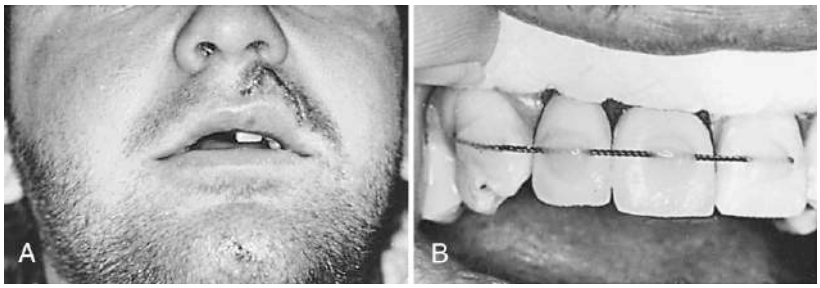


Figure 25.4 Splinting avulsed teeth. (A) Patient with traumatically avulsed maxillary right incisors. (B) Completed splint stabilizes repositioned incisors.

III. porcelain-fused-to-metal pontic or all-metal pontic with metal retainers and
IV. all-porcelain pontic.

Although the four types differ in the degree of permanency, they share a major advantage—conservation of the natural tooth structure. In addition, they can be viable alternatives to the conventional fixed bridges in circumstances where age, expense and clinical impracticality are considerations.

Because of the conservative preparation and bonded nature of all of these bridge types, retention is never as strong as in the case of a conventional bridge. As part of informed consent, patients should be told of the risk, although remote, of swallowing or aspirating bonded bridges that are dislodged. To reduce the risk of dislodgment, patients should be cautioned not to bite hard foods or objects with bonded bridge pontics.

The ideal site for a conservative bridge is where the edentulous space is no wider than one or two teeth. Other considerations include bite relation, oral hygiene, periodontal condition and extent of caries, defects and restorations in the abutment teeth. Conservative bridges are especially indicated for young patients because the teeth usually have large pulp chambers and short clinical crowns. Many older patients with gingival recession and mobile teeth are prime candidates because splinting can be incorporated with the bridge. More specific indications and clinical procedures for each of the four types of bridges are presented in the following sections.

I. Natural Tooth Pontic

The crowns of natural teeth (primarily incisors) often can be used as acid-etched, resin-bonded pontics. Considerations for this type of treatment include the following:

- periodontally involved teeth warrant extraction,
- teeth have fractured roots,
- teeth are unsuccessfully reimplanted after avulsion and
- root canal treatment has been unsuccessful.

However lost, the immediate replacement of a natural anterior tooth has a great psychologic value for most patients, although the procedure may be temporary. Natural tooth pontics also can be placed as interim restorations until an extraction site heals if conditions require a conventional bridge or an implant.

Certain prerequisites must exist to ensure a successful result:

I. the extracted tooth and abutments must be in reasonably good condition, especially the pontic, because it may become brittle and more susceptible to fracture;
II. the abutment teeth should be fairly stable and
III. the pontic must not participate in heavy centric or functional occlusion.

Because of this third restriction, canines and posterior teeth are not usually good candidates for this procedure. If the adjacent teeth are mobile, it is frequently necessary to secure them by splinting with composite (see the section on Techniques for Splinting Anterior Teeth).

Technique

- A maxillary right central incisor is being extracted for periodontal reasons (Fig. 25.5A and B). Before the tooth is extracted, a small round bur is used to place a shallow identifying mark on the facial surface to indicate the level of the gingival crest.
- If the tooth to be extracted is well-positioned in the dental arch, a PVS (Polyvinyl Siloxane) bite registration or putty impression material can be used to generate an index to reposition the natural tooth pontic in the correct pre-extraction position. After extraction, a 5 cm × 5 cm (2 in. × 2 in.) sponge is held in the space with pressure for haemorrhage control.
- By using a separating disk or a diamond instrument, the extracted tooth is transversely cut a few millimetres apical to the identification mark. When pontic length is determined, shrinkage of the healing tissue underlying the pontic tip must be anticipated. The root end is discarded.
- If the pulp canal and chamber have completely calcified, the next procedure is shaping and polishing the apical end of the natural tooth pontic as described in the following paragraphs. If the chamber is calcified as disclosed on the radiograph and the canal is nearly calcified, the canal is opened from the apical end by using a small round bur or diamond to the extent of the canal.
- The operator should be as conservative of the tooth structure as possible and yet provide access for subsequent injection of the composite material to fill the canal. A large chamber and canal are instrumented and debrided using conventional endodontic procedures with access from the apical end (Fig. 25.5C).
- Access is provided for subsequent injection of composite. Removal of the pulpal tissue in this manner prevents discolouration of the tooth caused by degeneration products. Traditional lingual access for instrumentation is avoided to prevent weakening the pontic. After these procedures, the canal (and chamber, if present) is filled and closed with composite.
- After composite has been fully polymerized, the apical end is contoured to produce a bullet-shaped ovate design (Fig. 25.5C). This design provides adaptation of the pontic tip to the residual ridge, and yet it allows the tissue side of the pontic tip to be cleaned with dental floss. It is also the *most esthetic pontic tip design* that can be used.
- While being contoured, the tip is occasionally evaluated by trying the pontic in the space. In the maxillary



Figure 25.5 Resin-bonded maxillary natural tooth pontic. (A) Preoperative photograph before extraction of periodontally involved maxillary right central incisor. (B) Extraction site immediately after the removal of an incisor. (C) Enlarged apical opening ready to be filled with composite. The pontic tip has been contoured to an ovate design. (D) The abutment teeth are isolated, roughened and acid etched. (E) Immediate postoperative photograph of natural tooth pontic bonded in place. (F) Resin-bonded natural tooth pontic with healed residual ridge 6 weeks later.

arch, passive contact between the pontic tip and the healed residual ridge is considered ideal for maximal phonetic and esthetic potential. In the mandibular arch (where esthetics is not generally a problem), the pontic tip is best shaped into the same bullet-shaped design but positioned as a hygienic pontic type that does not contact tissue (Fig. 25.6A). The pontic tip is smoothed and polished using a proper sequence of abrasive disks or polishing points. A polished pontic tip not only is easier to clean but also retains less plaque.

- Usually, a rubber dam is needed for isolation of the region to prevent seepage of blood and saliva. Isolation using cotton rolls and gingival retraction cords is acceptable if the haemorrhage has been controlled. Any caries lesions or faulty proximal restorations on involved proximal surfaces of the pontic and the abutments are restored with light-cured composite (preferably the same material to be used subsequently for the bridge connectors) by using modified preparation designs. It is recommended that the resulting restored surfaces be under-contoured rather than over-contoured to facilitate positioning of the natural tooth pontic.
- Next, the involved proximal surfaces on the abutment teeth and the pontic are roughened with a coarse, flame-shaped diamond instrument. Spaces of approximately 0.5 mm should exist between the pontic and the abutment teeth, because stronger connectors are provided by the additional bulk of the composite material. Now, the operator should acid-etch, rinse and dry all the prepared (i.e. roughened) surfaces (Fig. 25.5D).
- Light-cured composite is preferred for bonding natural tooth pontics because the extended working time allows the operator to contour the connectors before polymerization. First, the adhesive is applied to the etched surfaces of the pontic and lightly blown with air to remove the excess. Then it is polymerized by application of light, and the pontic is set aside (ready for bonding in the mouth). Next, the adhesive is applied to the etched surfaces of the abutment teeth and cured. A small amount of composite material is placed on the proximal contact areas of the natural tooth pontic, and the pontic is inserted carefully in the proper position in the mouth. The composite is shaped around the

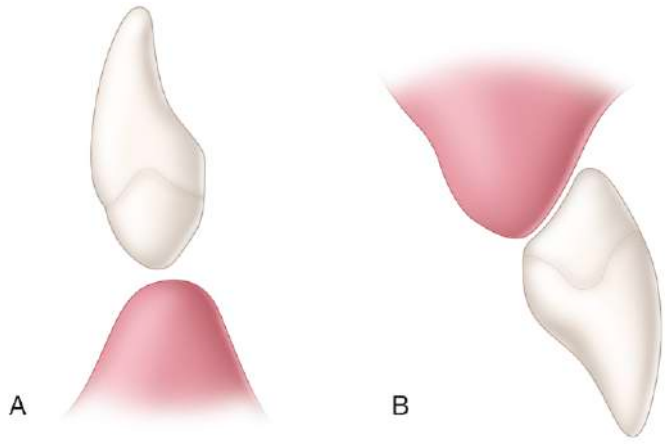


Figure 25.6 Pontic tip design. (A) Hygienic-type pontic with ovate or bullet-shaped tip. (B) Modified ridge lap-type pontic with slight concavity conforming to residual ridge.

contact areas with an explorer tip. After final verification that the pontic position is correct, composite is polymerized with light. Next, additional composite is applied in the proximal areas (more material is added on the lingual than on the facial surface), contoured and cured. Adequate gingival embrasures must be provided to facilitate flossing and ensure gingival health. After sufficient material has been added and polymerized, the embrasure areas should be shaped and smoothed with carbide finishing burs or fine diamonds and polishing disks or points.

- The rubber dam is removed, and the occlusion is evaluated for centric contacts and functional movements. Heavy contacts on the pontic or the connector areas must be adjusted. The finished bridge immediately after bonding is illustrated in Fig. 25.5E.
- The patient should return in 4–6 weeks for evaluation of the relationship of the pontic tip to the tissue. Passive contact should exist between the pontic tip and the underlying tissue to prevent ulceration. If tissue ulceration is present, the pontic must be removed, recontoured and rebonded. The finished bridge and healed residual ridge are shown in Fig. 25.5F.

As stated earlier, abutment teeth that are mobile often can be splinted with composite to afford stability to periodontally involved teeth. The abutments are isolated, roughened and acid-etched (Fig. 25.7A). Because esthetics is not as crucial, a hygienic pontic tip is recommended for mandibular incisors (Fig. 25.6A). The finished bridge splint is illustrated in Fig. 25.7B.

II. Denture Tooth Pontic

An acrylic resin denture tooth can be used as a pontic for the replacement of missing maxillary or mandibular incisors by using the resin-bonding technique (Fig. 25.8A–H).

Although this type of bridge is sometimes used as an interim prosthesis and is called a *temporary bridge*, it can be a viable alternative to a conventional bridge and may last for years in some circumstances.

As with the natural tooth pontic, the *major contraindications* to this type of resin-bonded bridge are:

I. abutment teeth that have extensive caries or restorations,
II. abutment teeth that have mobility and
III. a pontic area that is subjected to heavy occlusal forces.

In the illustrated example, the permanent maxillary right lateral incisor is missing and the adjacent teeth are in favourable condition and position (Fig. 25.8A). Further examination reveals an ideal situation for a conservative bridge that uses a denture tooth pontic.

Technique

- Although the entire procedure can be completed at chairside in one appointment, considerable time can be saved by an indirect technique. During the first appointment, the shade (Fig. 25.8) and mold of the denture tooth are selected, and impressions are made.
- In the laboratory, stone casts are poured, and the ridge area is relieved slightly and marked with a soft lead pencil. As the pontic is trial positioned, the pencil markings rub off onto its tip to facilitate contouring of this area (Fig. 25.8C).
- Contouring is best accomplished with acrylic burs and a Burlew wheel in a straight handpiece. The tissue side of the pontic should be contoured to a modified ridge lap configuration that is convex mesiodistally and slightly concave faciolingually (Fig. 25.6B).
- This type of design not only allows the pontic tip to adapt to the residual ridge, but it also allows for effective cleaning with dental floss. After it is contoured, the pontic tip should be smoothed and highly polished with pumice and an acrylic-polishing agent (Fig. 25.8D).
- Because composite does not normally bond to acrylic resin, provisions must be made to facilitate a strong connection between the pontic and the adjacent teeth. One provision may be completed in the laboratory by preparing large Class III conventional preparations in the pontic that mechanically retain the composite material. The outline of the preparations must be large

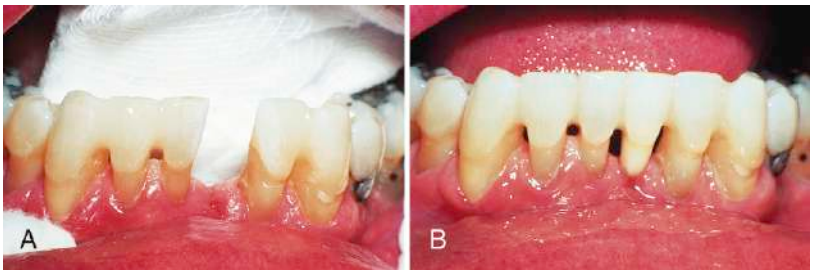


Figure 25.7 Resin-bonded mandibular bridge splint using natural tooth pontic. (A) The anterior segment is splinted with composite, and the abutment teeth are isolated, roughened and etched. (B) Natural tooth pontic is bonded in place.

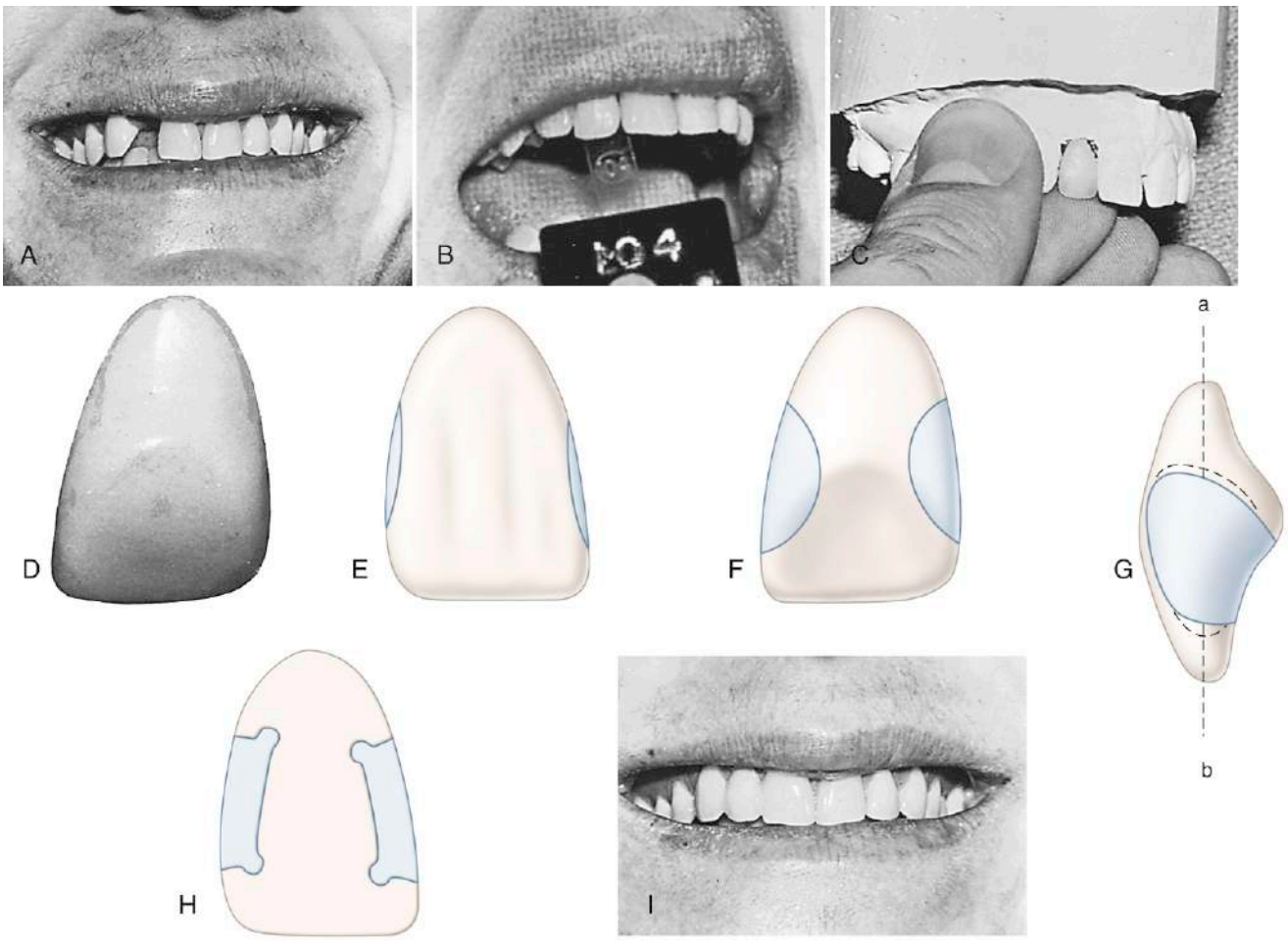


Figure 25.8 Resin-bonded denture tooth pontic. (A) Preoperative photograph shows a missing maxillary lateral incisor. (B) Shade and mold selection. (C) Positioning pontic on working model while contouring. (D) Contoured and polished pontic (lingual view). (E–G) Outline form of Class III preparations: facial (E), lingual (F) and proximal (G) views. (H) Cross section of denture tooth (longitudinal section) in plane *ab* as seen in (G) showing the mechanical retention form incisally and gingivally as prepared with a No. ½ bur. (I) Denture tooth pontic is bonded in place with composite.

enough to provide adequate surface area of the composite restoration for bonding to the adjacent teeth (Fig. 25.8E–G).

- An appropriately sized round bur (No. 2 or No. 4) is used to cut each preparation to a depth of approximately 1.5 mm and extend the outline approximately 0.5 mm past the contact areas into the gingival, incisal and facial embrasures. Even more extension should be made into the lingual embrasure to provide for bulk of composite material in the connector areas.
- The lingual extensions should not be connected because this unnecessary step would unduly weaken the pontic. Mechanical undercuts are placed at the incisoaxial and gingivoaxial line angles with a No. ½ bur to lock the composite material (to be inserted later in the technique) mechanically in the acrylic resin pontic (Fig. 25.8G and H).
- At the next appointment, the pontic is tried in place to confirm that the shade and contours are correct. Approximately 0.5 mm of space should exist between each proximal 'contact' and the abutment tooth. The pontic is cleaned with acetone to remove dust and debris.
- Retention of the pontic by undercuts, as previously described, also can be augmented by a second provision – the conditioning of the proximal aspects of the pontic with two applications of ethyl acetate, a polymer softener. A thin layer is applied in the Class III preparations and on the cavosurface areas and allowed to dry for 5 min. This process is repeated to ensure optimal bonding.
- The preparations are filled with the same light-cured composite material expected to be used for bonding the pontic in place. The composite should be applied and cured in the retentive areas before the remainder of the preparation is filled. This step ensures complete polymerization. After the entire preparation is filled, it should be polymerized again with the light source. It is better to leave the contact areas slightly under-contoured for the pontic to fit easily between the abutment

teeth. The pontic is set aside in a safe place for some time.

- Isolation of the abutment teeth should be accomplished with cotton rolls and retraction cords (rather than with a rubber dam) to relate the pontic better to the residual ridge area. Any caries or old restorations in the adjoining proximal areas of the abutment teeth should be removed at this time, and any indicated liners should be applied. The proximal surfaces of the abutment teeth are roughened with a coarse flame-shaped diamond instrument. This step is followed by acid-etching, rinsing and drying. The adhesive is applied, lightly blown with air and cured. Tooth preparations, if present, are restored with the same composite material. Care is taken not to over-contour the restoration or restorations.
- The pontic is evaluated by positioning it temporarily in the edentulous space. If adjustments are made, the surfaces should be cleaned with acetone. Next, a small amount of composite is wiped onto the contact areas (mesial and distal) of the pontic, and the pontic is placed into the proper position between the abutment teeth. An explorer tip is helpful in placing the material evenly around the contact area. Care must be taken to place the pontic so that it lightly touches the ridge, but does not cause tissue blanching. The composite material used to position the pontic is polymerized.
- It is helpful to add and cure the additional composite in small increments to obtain the correct contour and minimize finishing procedures. The facial, incisal and gingival embrasures should be defined with a flame-shaped finishing bur or fine diamond and polished with appropriate disks or points. The lingual aspect of the bridge is contoured with a round finishing bur without defining lingual embrasures because this could weaken the connectors. The retraction cords are removed from the gingival crevice. Articulating paper is used to mark the occlusion, and any offensive contacts are removed. The final restoration is shown in Fig. 25.8I.

III. Porcelain-fused-to-metal Pontic or All-metal Pontic with Metal Retainers

A stronger and more permanent type of acid-etched, resin-bonded bridge is possible by use of a cast metal framework.[3,4]

- In anterior areas where esthetics is a consideration, the design of the bridge includes a porcelain-fused-to-metal (PFM) pontic with metal winged retainers extending mesially and distally for attachment to the proximal and lingual surfaces of the abutment teeth.
- In posterior areas where esthetics is not a critical factor, the bridge can have either a PFM or an all-metal pontic.

The technique is more complicated and time consuming than the previously described methods because it requires some initial tooth preparation, an impression, laboratory procedures and a second appointment for etching and bonding. Compared with conventional bridges, resin-bonded bridges of this type offer five distinct advantages:

I. Anaesthesia is usually not required.
II. The tooth structure is conserved (i.e. no dentin involvement).
III. Gingival tissues are not irritated because margins usually are not placed sub-gingivally.
IV. An esthetically pleasing result can be obtained more easily.
V. The cost is lower because less chair time is required, and laboratory fees are lower as well.

Ideally, this type of conservative bridge is used for short spans in the anterior or posterior areas with sound abutment teeth in good alignment. The most favourable occlusal relationship exists where little or no centric contact and only light functional contact are present. However, teeth can be prepared and the bridge framework designed to withstand moderately heavy occlusal forces. Orthodontics may be required to improve tooth alignment. The bridge also can be extended to splint adjacent periodontally involved teeth. Surgical crown-lengthening procedures sometimes are indicated for teeth with short clinical crowns.

Although minimal, some preparation of the enamel of the abutment teeth is mandatory in the retainer area of the bridge to:

- provide a definite path of insertion or seating or both,
- enhance retention and resistance forms,
- allow for the thickness of the metal retainers and
- provide physiologic contour to the final restoration.

The importance of the tooth preparation design cannot be overemphasized. The success of these types of bridges depends on the preparation design. The bridges must be independently retentive by design and cannot rely solely on resin bonding for retention. Preparation design for these types of bridges is similar to that for a cast three quarter crown; however, it is restricted to enamel.

The preparation for each abutment varies, depending on the individual tooth position and anatomy. Approximately the same amount of surface area should be covered on each abutment tooth. In some situations, recontouring of the adjacent and opposing teeth may be indicated. The details of the preparations are described later.

Two primary types of resin-bonded bridges with metal retainers currently exist[3,4]:

i. Rochette Resin-bonded Bridges with Metal Retainers

The Rochette type uses small countersunk perforations in the retainer sections for retention and is best suited for anterior bridges (Fig. 25.9A).[4] Care must be exercised in placing the perforations to prevent weakening the framework. Perforations that are too large or too closely spaced invite failure of the metal retainer by fracture. The perforations

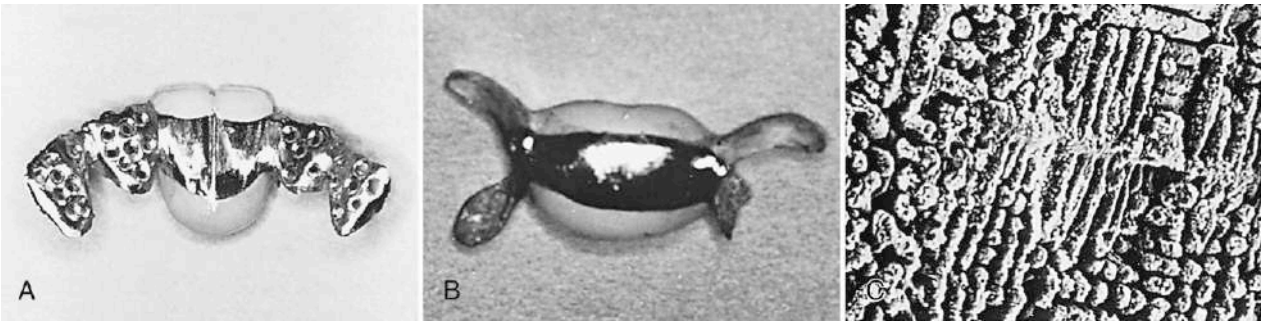


Figure 25.9 Acid-etched, resin-bonded metal bridges. (A) Rochette type. (B) Maryland type. (C) Scanning electron micrograph of etched metal surface.
(Courtesy: Dr John Sturdevant.)

should be approximately 1.5 to 2 mm apart and have a maximum diameter of 1.5 mm on the tooth side. Each hole is countersunk so that the widest diameter is towards the outside of the retainer. When the bridge is bonded with a resin cement, it is mechanically locked in place by microscopic undercuts in the etched enamel and the countersunk holes in the retainer (Fig. 25.10A).

The advantages of this design include the following:

- It is easy to see the retentive perforations in the metal.
- If the bridge must be removed or replaced, the bonding medium can be cut away in the perforations to facilitate easy removal.
- No metal etching is required.

The disadvantages of this design include the following:

- The perforations, if improperly sized or spaced, could weaken the retainers.
- The exposed resin cement is subject to wear.
- It is not possible to place perforations in proximal or rest areas.

ii. Maryland Resin-bonded Bridges with Metal Retainers

A second type of cast metal framework, commonly known as the Maryland bridge, is reported to have improved bonding strength (Fig. 25.9B).[3,5] Instead of perforations, the *tooth side of the metal framework is electrolytically or chemically etched*, which produces microscopic undercuts (Fig. 25.9C).

The bridge is attached with a self-cured, resin-bonding medium that locks into the microscopic undercuts of the etched retainer and the etched enamel (Fig. 25.10B). It can be used for anterior and posterior bridges. Although this design has been reported to be stronger, it is more technique sensitive because the retainers may not be properly etched or may be contaminated before cementation. Because the retentive features cannot be seen with the unaided eye, the etched metal surfaces must be examined under a microscope to verify proper etching (minimum magnification).

More recently, Maryland bridges have been fabricated with no electrolytic etching of the surface and chemically bonded to the tooth after a process called *silicoating* or with a 4-META or phosphate ester-containing, resin-bonding medium.[6,7] Resin materials containing 4-META or other resin monomers are capable of strongly bonding to metal surfaces.[8,9] Surface roughening with micro-etching (i.e. sandblasting) is commonly used in conjunction with these adhesive cements. These types of Maryland bridges are referred to as *adhesion bridges* and differ only in the means of retention. The design of adhesion bridges is the same for this alternative Maryland bridge design.

Successes and failures have been observed with both bonded bridge designs. Because the procedures are technique-sensitive, every step must be followed carefully.

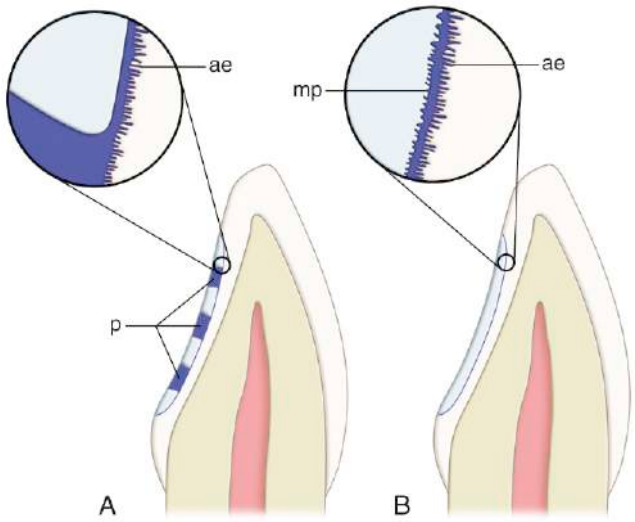


Figure 25.10 Cross-sectional diagram of two types of resin-bonded bridges. (A) In addition to acid-etching prepared enamel surfaces (*ae*), the Rochette type uses small countersunk perforations (*p*) in the retainer section. (B) In the Maryland type, the tooth side of the framework is either etched to produce microscopic pores (*mp*) or bonded with no etching with an adhesive cement.

IV. All-porcelain Pontic

Improvements in dental porcelains along with the capacity to etch and bond strongly to porcelain surfaces have made all-porcelain pontics a viable alternative to pontics with metal winged retainers (e.g. Maryland and Rochette bridges).[10,11]

- Although all-porcelain pontics are not as strong as pontics with metal retainers, far superior esthetic results can be achieved because no metal substructure or framework is present.
- All-porcelain pontics often can be used when tooth anatomy precludes or restricts the preparation and placement of a metal winged pontic. Long, pointed canines with proximal surfaces exhibiting little occlusogingival height often lack adequate areas for the placement of retention grooves.
- Anterior teeth that are notably thin faciolingually also are not good candidates for metal, resin-bonded bridge retainers and often are esthetic failures because of metal showing through the tooth. In such instances, custom-fabricated, etched porcelain pontics frequently can provide an esthetic, functional alternative.
- All-porcelain pontics are particularly indicated in adolescents and young adults, in whom virgin, unrestored teeth are often encountered. Because teeth are not extensively prepared, this procedure is almost entirely reversible. This is a major benefit in young patients, where all-porcelain pontics can be placed as interim restorations until implants or a more permanent prosthesis can be placed at an older age.

Because of their limited strength, all-porcelain pontics should be considered provisional in nature, similar to the natural tooth pontic and the acrylic denture tooth pontic. Similar to the natural tooth and denture tooth pontics, certain prerequisites must be met to ensure a successful result:

I. First, the abutment teeth must be in reasonably good condition with proximal enamel surfaces that are intact or contain very small composite restorations.
II. Second, the abutment teeth should be stable with little mobility present. If the abutment teeth are mobile, it is frequently necessary to secure them as well by splinting with composite to adjacent teeth before placement of the bonded pontic (see section Techniques for Splinting Anterior Teeth).
III. Third, the pontic must not be placed in a position that would subject it to heavy centric or functional occlusal contacts. Because of these occlusion concerns, canines and posterior teeth are not usually good candidates for these types of resin-bonded bridges.

Technique

Fig. 25.11A and B illustrates a typical case of congenitally missing lateral incisors in which tooth contours contraindicated the use of resin-retained bridges with metal retainers. Central incisors are very translucent, and the mesial contours of canines are deficient (Fig. 25.11C and E). After assessing centric and functional occlusions, it was determined that all-porcelain pontics could be placed without subjecting them to heavy occlusal forces.

- At the first appointment, the involved abutments are cleaned with flour of pumice and an accurate shade selection is made, noting any desired colour gradients or characterizations.
- No preparation of the teeth is recommended unless the proximal surfaces of the abutment teeth adjacent to the edentulous space are markedly convex. In such cases, slight flattening of the proximal surfaces with a diamond instrument facilitates closer adaptation of the pontic to the abutment teeth, increasing strength of the connectors. Otherwise, no retentive features are recommended for the preparation in the abutment teeth; the connector areas are entirely made of composite.
- Bridge connectors composed of porcelain are subject to eventual fatigue fracture, after which repair is made more difficult. Studies show that *'veneer bridges'* (i.e. all-porcelain pontics retained by adjacent etched porcelain veneers) in particular are the weakest design of all and should be avoided.[11] These types of bridges not only provide little bond strength to the pontic but also needlessly cover adjacent, healthy facial tooth surfaces. All-porcelain pontics (composite used for bonding to the abutment teeth) are similar to extracted natural tooth pontics in this regard. Those that have connector areas consisting of the design feature allow for easy repair and replacement of the composite connector should a fracture in this area be encountered.
- If high-strength ceramics that are totally immune to crack propagation and cohesive fracture are developed, retentive features prepared in the adjacent abutment teeth may be desired. These features, prepared in enamel, would consist of *proximal grooves* or *boxes*, depending on the faciolingual dimension of the proximal surfaces. In the absence of such totally fracture-resistant ceramics, however, all-porcelain pontics are best placed with composite connectors for ease of repair and replacement.
- An elastomeric impression is made, and a working cast is generated from it. A modified ridge lap pontic tip design (Fig. 25.6B) is recommended. An occlusal bite registration should be made and forwarded to the laboratory so that the occlusal relationship can be considered during fabrication of all-porcelain pontics.
- The proximal surfaces of the pontics are etched with hydrofluoric acid. The area etched must include all areas anticipated for bonding to the composite-bonding medium. The etched proximal surfaces should extend just beyond the lingual line angles so that additional



Figure 25.11 All-porcelain pontics. (A and B) Patient with congenitally missing lateral incisors. (C and D) Right side before and after treatment. (E and F) Left side before and after placement of all-porcelain pontic. (G) Lingual view of completed bridges. (H) Facial view of all-porcelain pontics.

composite can be placed in the lingual embrasure areas for additional connector strength.

- At the subsequent appointment, teeth are isolated with cotton rolls. A 5 cm × 5 cm (2 in. × 2 in.) cotton gauze is placed across the back of the patient's mouth to act as a protective shield should the pontic be inadvertently dropped. A rubber dam is not recommended for this procedure because it precludes accurate assessment of the adaptation of the pontic tip to the residual ridge.
- Before the teeth dehydrate, the position of each pontic is tested in the edentulous space to assess the shade and relationship of the pontic tip to the residual ridge. The pontic tip should contact the residual ridge passively with no blanching of the underlying tissue evident. Spaces of approximately 0.3–0.5 mm should exist between the pontic and the abutment teeth because stronger connectors are provided by the additional bulk of composite material.
- Care must be taken not to allow contamination of the etched pontic from saliva to occur during the try-in phase. If saliva contamination occurs, the etched proximal surfaces of the pontic must be cleaned thoroughly with alcohol and dried. After try-in, all etched proximal surfaces of the porcelain pontics are primed with a suitable silane-coupling agent (see the manufacturer's instructions for the specific technique). The pontics are now ready for bonding.
- The involved proximal enamel surfaces of the abutment teeth are roughened with a coarse, flame-shaped diamond instrument. Thereafter, all of the prepared (i.e. roughened) enamel surfaces should be acid etched, rinsed and dried. Care must be taken to maintain clean, dry, uncontaminated etched surfaces until the pontic is positioned and bonded. The abutment teeth are now ready for bonding.
- A light-cured composite is preferred for bonding all-porcelain pontics because the extended working time allows the operator to contour the connectors initially before polymerization. The dentist applies the adhesive to the etched surfaces of the porcelain pontic and the abutment teeth and lightly blows with air to remove the excess. A 20-s application of light from the light-curing unit is used to polymerize the bonding agent on each etched surface.
- A small amount of composite material is placed on the proximal contact areas of the natural tooth pontic and the pontic is inserted carefully into the proper position in the edentulous space. A stent, or index, made from bite registration material or fast-setting plaster can be used to position the pontic, if desired. Positioning by hand is recommended, however, so that optimal gingival pressure can be maintained for best tissue adaptation.
- The dentist shapes the excess composite extruding from the connector areas around the contact areas with an explorer tip or small plugger end of a composite instrument. After final verification that the pontic position is correct, the composite is polymerized with light for a minimum of 40–60 s each from facial and lingual directions (for a total of 80–120 s).
- Additional composite is applied in the proximal areas (more material is added on the lingual surface than on the facial surface), contoured and polymerized. Adequate gingival embrasures must be maintained to facilitate flossing and ensure good gingival health. After sufficient material has been added and polymerized, the dentist shapes and smoothes the embrasure areas with carbide finishing burs, fine diamonds and polishing disks. Facial embrasures are defined for esthetics, but lingual embrasures are closed with composite to strengthen the connectors (Fig. 25.11D, F and G).
- The dentist evaluates the occlusion centric contacts and functional movements. Heavy contacts on the pontic or the connector areas must be adjusted. The finished bridges (immediately after bonding) are illustrated in Fig. 25.11D and F–H. As with all resin-bonded bridges, patients must be advised to avoid biting into hard foods or objects to reduce the risk for dislodgment.

Also, the patient must be advised as part of informed consent that although the chances are remote, the potential for dislodgment and the risk of swallowing or aspirating the pontic do exist. This possibility exists for all resin-bonded bridges, and patients must be warned of this hazard even though the risk is minimal.

Summary

Stabilization of mobile teeth due to trauma, periodontal disease or after fixed orthodontic treatment using a resin-bonded splint is an important clinical procedure that a clinician should master. The use of a resin-bonded bridge should not be the primary treatment of choice for missing teeth; however, this offers a conservative and esthetic alternative in certain clinical situations.

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